# Experiment-4

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**Course Code: -**MLA0305 **Slot-**B

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**Title**

**Implementation of Multi-Armed Bandit Problem using Exploration and Exploitation**

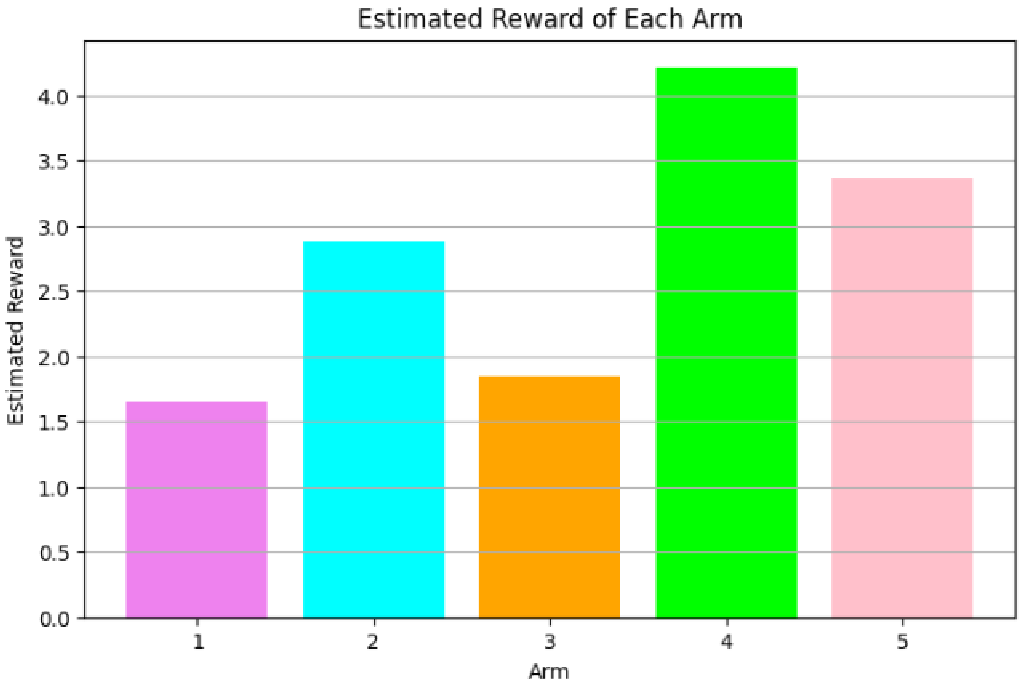
**Aim**

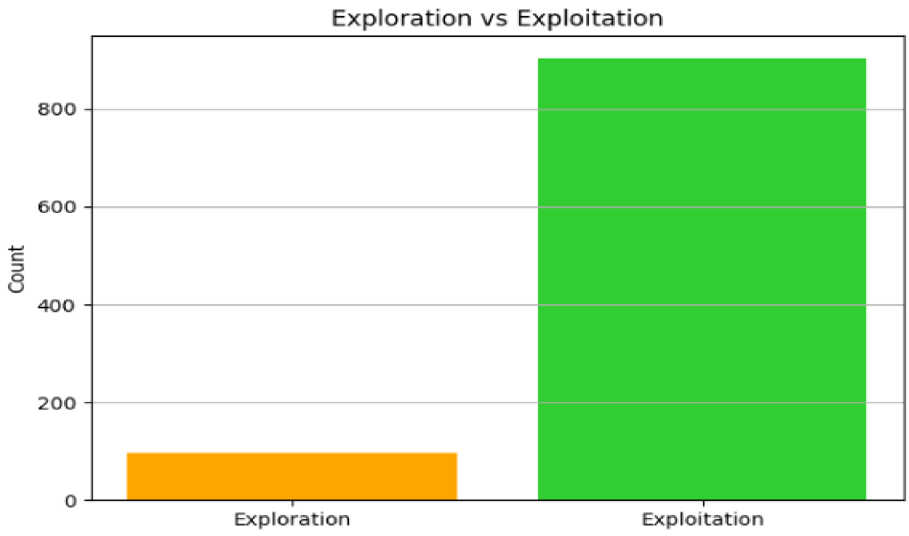
To implement the Multi-Armed Bandit problem using the Epsilon-Greedy algorithm and study the trade-off between exploration and exploitation while identifying the arm with the highest expected reward.

**Procedure**

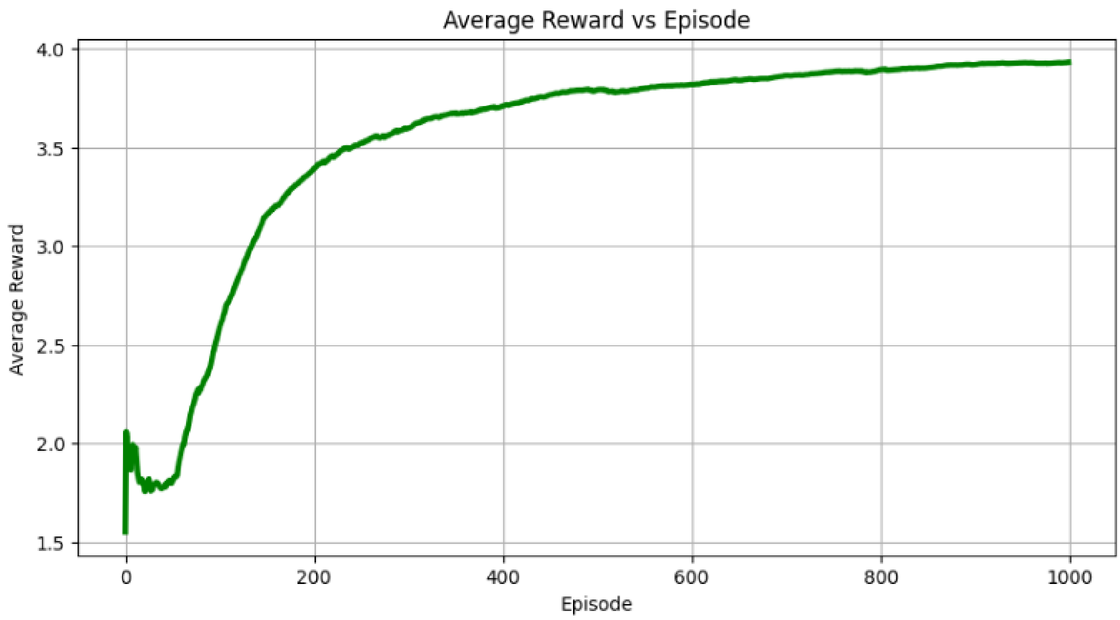
1. Define a Multi-Armed Bandit environment with multiple arms.
2. Assign a true reward distribution to each arm.
3. Initialize the estimated reward of every arm to zero.
4. Set the number of episodes and the value of epsilon (ε).
5. At each episode, generate a random number.
6. If the random number is less than ε, select a random arm (exploration).
7. Otherwise, select the arm with the highest estimated reward (exploitation).
8. Generate a reward from the selected arm.
9. Update the estimated reward of the selected arm using the observed reward.
10. Record the number of times each arm is selected.
11. Repeat the process for the specified number of episodes.
12. Calculate the exploration and exploitation counts and percentages.
13. Identify the arm having the highest estimated reward.
14. Compare the true reward and estimated reward of each arm.
15. Display the results using summary tables and graphs.
16. Analyze how the agent balances exploration and exploitation.

## Visualizations

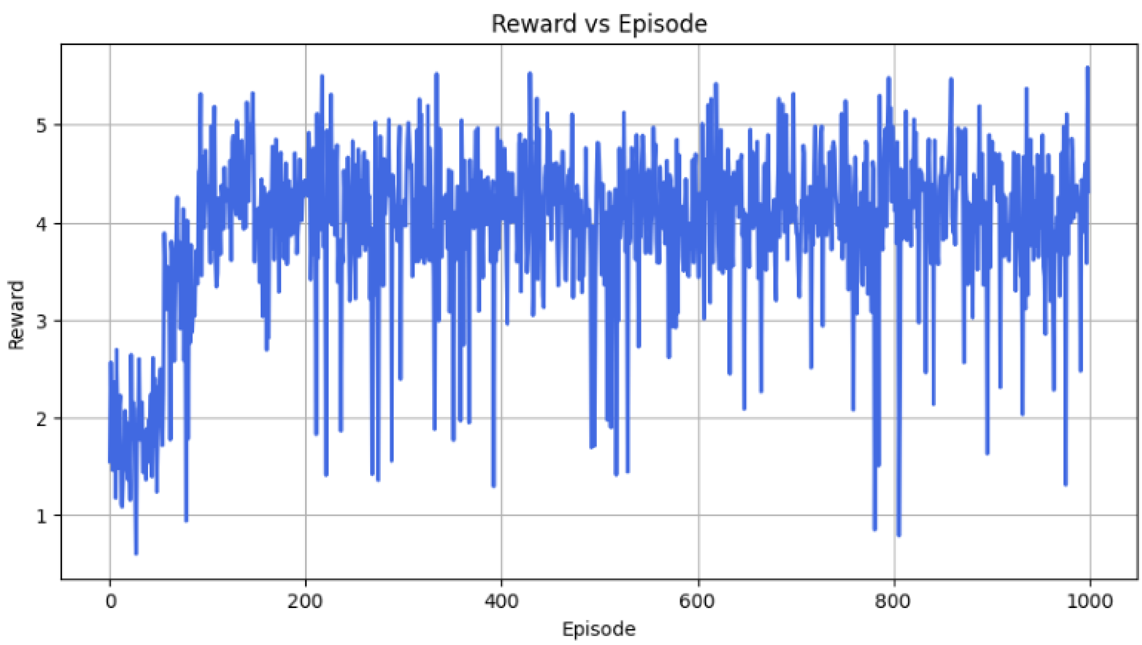





Exploration vs Exploitation
Count
0
200
400
600
800
Exploration
Exploitation

Average Reward vs Episode
Average Reward
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
Episode

Reward vs Episode
Reward
1
2
3
4
5
0
200
400
600
800
1000
Episode

**Summary Table**

| Arm | True Reward | Estimated Reward | Times Selected | Selection % |
|---|---|---|---|---|
| 1 | 1.5 | Obtained from program | Obtained | Obtained |
| 2 | 2.8 | Obtained from program | Obtained | Obtained |
| 3 | 1.9 | Obtained from program | Obtained | Obtained |
| 4 | 4.2 | Obtained from program | Obtained | Obtained |
| 5 | 3.4 | Obtained from program | Obtained | Obtained |

**Exploration vs Exploitation**

| Method | Count | Percentage |
|---|---|---|
| Exploration | Obtained | Obtained |
| Exploitation | Obtained | Obtained |
| Total | 1000 | 100% |

**Result**

The Multi-Armed Bandit problem was successfully implemented using the Epsilon-Greedy algorithm. The agent performed exploration by trying different arms and exploitation by selecting the arm with the highest estimated reward. With repeated interactions, the estimated rewards approach the true rewards and the agent identifies the best-performing arm.